

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks Available |          |          |           |          |           |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|-----------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                 | AO1             | AO2      | AO3      | Total     | Maths    | Prac      |
| 3.       | (a) |       |  | Order is: B, C, A, D.<br>3 marks for all 4 correct<br>2 marks for 2 or 3 correct<br>1 mark for 1 correct                                                                                                                                                                                                                                                                                                                                        | 3               |          |          | 3         |          | 3         |
|          | (b) |       |  | <b>Any 2 × (1) from:</b><br>- Move the ruler nearer [the spring]<br>- Clamp the ruler in place / ensure the ruler is vertical<br>- Use a ruler marked in mm <b>or</b> with better resolution. Accept measure in mm.<br>- Use a pointer [from the (bottom of the) spring to the ruler] Don't accept use a laser pointer.<br>Don't accept repeat the experiment <b>or</b> use a metre ruler <b>or</b> use a camera <b>or</b> more accurate ruler. |                 |          | 2        | 2         |          | 2         |
|          | (c) | (i)   |  | 4 [N]                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1               |          |          | 1         | 1        | 1         |
|          |     | (ii)  |  | $k = \frac{4 \text{ ecf}}{8} (1)$<br>= 0.5 [N/cm] (1)                                                                                                                                                                                                                                                                                                                                                                                           | 1               | 1        |          | 2         | 2        | 2         |
|          |     | (iii) |  | 400 g to 200 g (1)<br>So 8 cm goes to 4 cm (1) [so the claim is false].<br><b>Alternative:</b><br>Mass is half of 400 g (1)<br>So the extension is 4 cm (1) [so the claim is false].<br><b>Alternative:</b><br>2 cm is a quarter of the 8 cm extension (1)<br>so the force is 1 N / mass of 100 g (1) [so the claim is false].                                                                                                                  |                 |          | 2        | 2         | 1        | 2         |
|          |     |       |  | <b>Question 3 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>5</b>        | <b>1</b> | <b>4</b> | <b>10</b> | <b>4</b> | <b>10</b> |